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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2022-2023

Program: B.TECH | Year: III Semester: VI

Stream : Cyber Security

Subject: Digital Forensic and Incident Response | Time: 3 hrs (2:00 pm to 5:00 pm)

Date: 20/04/2023 | No. of Pages : 2

Marks: 100

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-1 ; SO-2 ; BL-2	a.	What are the characteristics of an Expert report?	5
CO-1 ; SO-2 ; BL-2	b.	What are the two main types of data hiding used in graphic files? Explain them.	5
CO-1 ; SO-2 ; BL-3	c.	A business impact analysis indicates an organization cannot operate without its web server for more than 5 days and still recover. The mean time to repair is 3 days. How many days do you have after a disaster to initiate repair or will the organization not be able to recover?	5
CO- 1; SO- 2; BL-2	d.	What is the difference between a logical address and a physical address in Microsoft file structures? Which type of address do sectors and clusters use?	5
Q2 CO-1 ; SO-2 ; BL-3	a	What are some of the direct types of evidence that you can collect and what are the indirect types of evidence in mobile forensic context? Which one of these can be a direct type of evidence in cyberstalking cases but indirect in other cases? Also, state means in a mobile device where information might be stored that should be retrieved for a forensics investigation.	10
CO-3 ; SO-2 ; BL-2	b	i) What is the significance of cyber law in our country? ii) State any five cybercrimes and respective applicable IPC for the same?	5 marks each

Q3 CO-1; SO2-; BL-2	a	Describe the steps for investigating routers and what are the various types of data you should collect?	10
CO-1; SO-2; BL-3	b	Explain the architecture of the Tor web browser. Which node on a Tor browser is exposed to the most risk and why?	10
Q4 CO-1; SO-2; BL-2	a	Describe the NTFS file system – what is the layout of an NTFS volume and what are the main components of it and what functions do they serve?	10
CO-3; SO-2; BL-2	b	Explain what CVSS is and how is it used in Incident Management Process. What are the various types of metrics used under this system?	10
Q5 CO-1; SO-2; BL-3	a	You're investigating a case involving a 2 GB drive that you need to copy at the scene. Write one to two paragraphs describing the three types of acquisitions—physical, logical, and sparse—you can use to copy the drive accurately.	10
CO-1; SO-2; BL-3	b	Several graphics files were transmitted via e-mail from an unknown source to a suspect in an ongoing investigation. The lead investigator gives you these graphics files and tells you that at least four messages are embedded in them. Explain the procedures you will follow to extract the hidden files from the graphic files and what tools can you use?	10
Q6 CO-1; SO-2; BL-2	a	Describe the structure of a report – what are the various sections and what should they contain? What are some of the factors to keep in mind to make the reports clear and precise?	10
CO-1; SO-2; BL-2	b	What is Denial of Service attacks? Describe at least 5 of them and how they exploit the TCP or UDP protocols.	10
Q7 CO-2; SO-2; BL-2	a	Describe the five phases of Incident Response Process and the functions of each phase	10
CO-3; SO-2; BL-2	b	What is the difference between a fact witness and an expert witness? Describe some of the ethical errors an expert witness should avoid.	10